

An Exact Computer-Assisted Optimality Certificate for Nine Disks of Radii $1, 2, \dots, 9$

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Abstract

We give an exact, independently replayable computer-assisted argument for the minimum radius of a circular container accommodating disks of radii $1, 2, \dots, 9$. The claimed optimum is the unique root of a six-angle closure equation near 19.23319390808612. An explicit construction proves attainability. The global lower bound reduces to an exhaustive rational radial tree with 47 nodes: 23 leaves reject all sixty cyclic orders using negative cycles in angle-difference constraints; the final leaf rejects fifty-nine orders by negative cycles and the remaining order by a strict radial monotonicity argument. The decisive replay checks use integer and rational arithmetic, not floating-point arithmetic. This is a computer-assisted draft, not a formal theorem-prover development or a peer-reviewed result.

1 Problem and exact statement

Let D_i be a disk of radius i with center $p_i \in \mathbb{R}^2$, $1 \leq i \leq 9$. For a container centered at the origin, feasibility at radius R means

$$\|p_i\| + i \leq R, \quad \|p_i - p_j\| \geq i + j \quad (i \neq j). \quad (1)$$

For $R > i + j$ define the angle between two wall-tangent, mutually tangent disks by

$$\beta_{ij}(R) = 2 \arcsin \sqrt{\frac{ij}{(R-i)(R-j)}}. \quad (2)$$

Let $C = (9, 5, 7, 6, 8, 4)$, with consecutive pairs interpreted cyclically, and set

$$\begin{aligned} H(R) = & \beta_{95}(R) + \beta_{57}(R) + \beta_{76}(R) \\ & + \beta_{68}(R) + \beta_{84}(R) + \beta_{49}(R) - 2\pi. \end{aligned} \quad (3)$$

Write R_0 for the unique zero of H in the rational interval

$$J_R = [19.23319390808612, 19.23319390808613]. \quad (4)$$

The exact root is strictly less than the rational tree cap

$$U = 19.23319390809. \quad (5)$$

Theorem 1.1 (Global optimality). *The minimum possible container radius for nine nonoverlapping disks of radii $1, 2, \dots, 9$ is exactly R_0 .*

The displayed decimals in (4) and (5) are terminating decimal fractions and hence exact rationals. They are not definitions by numerical approximation.

2 Exact upper bound: a nine-disk witness

Lemma 2.1 (Existence and uniqueness of the critical radius). *The function H has precisely one zero in J_R , and $R_0 < U$.*

Proof. All six angles are defined for $R \in J_R$, because $R > 17$ there. Their derivatives satisfy

$$\beta'_{ij}(R) = -\left(\frac{1}{R-i} + \frac{1}{R-j}\right) \tan \frac{\beta_{ij}(R)}{2} < 0.$$

An exact rational interval evaluation of square roots, arctangents, and π certifies $H(\inf J_R) > 0$ and $H(\sup J_R) < 0$; see the accompanying verifier `verify_nine_upper.py`. Continuity and strict monotonicity imply the assertion. The upper endpoint of J_R is smaller than U . $\square$

For each $i \in C$ set $s_i = R_0 - i$. Start with disk 9 at

$$p_9 = (R_0 - 9, 0).$$

Place the remaining core disks counterclockwise in the order 9, 5, 7, 6, 8, 4, with each successive angular gap equal to the corresponding $\beta_{ij}(R_0)$. Equation (3) closes this six-disk ring. All core disks touch the container and every cyclic neighboring pair is tangent.

Place the three other disk centers at the following *exact rational* coordinates:

$$p_1 = (-14.16, -11.20), \quad p_2 = (13.7125, -10.4353), \quad p_3 = (-3.94, 0.45). \quad (6)$$

The rational interval verifier evaluates the core centers over the full root interval J_R . It proves that the other $\binom{9}{2} - 6 = 30$ pairs have strictly positive squared nonoverlap slack, and that disks 1, 2, 3 lie strictly inside the container. The six core tangencies are consequences of the exact angular definitions and closure, not approximate numerical contacts. Therefore

$$R^* \leq R_0. \quad (7)$$

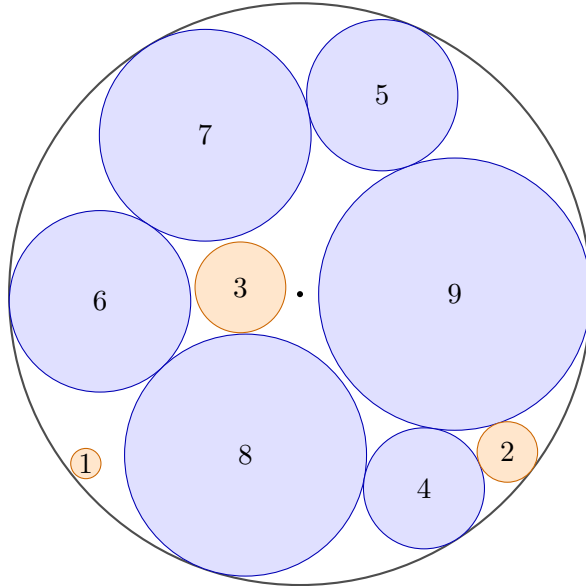


Figure 1: Illustrative nine-disk witness. The blue disks 4, 5, 6, 7, 8, 9 form a wall-tangent six-cycle. Coordinates in the illustration are rounded; the proof uses the exact angle-root construction and the rational auxiliary centers (6).

3 Angle constraints on radial boxes

We prove the lower bound for all placements of just the six largest disks $4, \dots, 9$. Let $s_i = \|p_i\|$. From containment and the triangle inequality, every packing with $R \leq U$ obeys

$$\ell_i = \max \left\{ 0, \max_{\substack{j \in \{4, \dots, 9\} \\ j \neq i}} (i + 2j - U) \right\} \leq s_i \leq U - i = u_i. \quad (8)$$

Every ℓ_i is positive. If a radial box is $s_i \in [\ell_i, u_i]$, the pairwise inequality $s_i + s_j \geq i + j$ permits sound propagation of its lower endpoints.

Lemma 3.1 (Four-corner radial-cosine bound). *For $d > 0$ and a rectangle $[a, b] \times [c, e] \subset (0, \infty)^2$, the function*

$$K_d(x, y) = \frac{x^2 + y^2 - d^2}{2xy}$$

attains its maximum at a corner.

Proof. For fixed y , $\partial K_d / \partial x = (x^2 - y^2 + d^2) / (2yx^2)$. Any interior critical point is a minimum, so the maximum over x is at an endpoint. The same argument on either resulting edge applies to y . $\square$

The law of cosines gives a necessary smaller angular separation between the two center rays:

$$\phi_{ij}(s_i, s_j) = \arccos K_{i+j}(s_i, s_j). \quad (9)$$

By Lemma 3.1, a proposed rational angular lower bound $q_{ij} \geq 0$ is sound whenever

$$C_{18}(q_{ij}) \geq \max_{x \in \{\ell_i, u_i\}, y \in \{\ell_j, u_j\}} K_{i+j}(x, y), \quad (10)$$

where C_{18} is the degree-18 Taylor polynomial for cosine, satisfying $C_{18}(q) \leq \cos q$ for $0 \leq q \leq \pi$. The verifier checks (10) exactly with rational arithmetic.

After orienting and lifting a cyclic order $(i_0, \dots, i_5)$ of the six center rays, every $a < b$ must satisfy

$$q_{i_a i_b} \leq \theta_b - \theta_a \leq 2\pi_+ - q_{i_a i_b}, \quad \pi < \pi_+ = 3.141592653590. \quad (11)$$

Lemma 3.2 (Negative-cycle exclusion). *A collection of necessary inequalities $\theta_v - \theta_u \leq w_{uv}$ is infeasible if its directed graph contains a cycle of strictly negative total weight.*

Proof. The inequalities telescope on a directed cycle to $0 \leq \sum w_{uv} < 0$. $\square$

4 The finite lower-bound certificate

A six-element labeled cyclic order has $5!/2 = 60$ representatives modulo rotation and reflection. We root each order at disk 9 and fix a representative orientation. Reflections preserve the radial constraints and distances, hence no configuration is lost.

The certificate begins at the full box (8). Each internal node bisects one radial interval at an exact rational cut, and the two closed children cover the parent. A leaf either excludes all 60 orders by explicit negative cycles in (11), or records negative cycles for 59 orders and applies the following analytic barrier to the last order.

Lemma 4.1 (Local wall-angle barrier). *Let $C = (9, 5, 7, 6, 8, 4)$. Suppose a radial box with lower endpoints ℓ_i has, for each successive pair i, j of C , the inequalities*

$$\begin{aligned} \ell_i + \ell_j &> i + j, \\ \max\{|\ell_i - (U - j)|, |(U - i) - \ell_j|\} &< i + j, \\ \ell_i^2 - (U - j)^2 + (i + j)^2 &> 0, \\ \ell_j^2 - (U - i)^2 + (i + j)^2 &> 0. \end{aligned}$$

If a packing in that box exists with container radius $17 < R < R_0$, then its six largest center rays cannot have cyclic order C or its reverse.

Proof. Differentiation of (9) gives, where $0 < \phi_{ij} < \pi$,

$$\frac{\partial \phi_{ij}}{\partial s_i} = -\frac{s_i^2 - s_j^2 + (i + j)^2}{2s_i^2 s_j \sin \phi_{ij}}.$$

The hypotheses guarantee that the derivative with respect to either radial coordinate is strictly negative throughout the extension of the box to $[\ell_i, U - i]$ for each disk. The strict triangle inequalities keep the angles well-defined throughout the comparison. Replacing the six radial coordinates by their wall values $R - i$ *only in the angle formulas*, without moving the physical disks, shows

$$2\pi \geq \sum_{(i,j) \in C} \phi_{ij}(s_i, s_j) \geq \sum_{(i,j) \in C} \beta_{ij}(R) > \sum_{(i,j) \in C} \beta_{ij}(R_0) = 2\pi,$$

a contradiction. The final strict inequality uses $R < R_0$ and the strict decrease of the functions β_{ij} in R . $\square$

Proposition 4.2 (Verified finite cover). *The accompanying exact rational checker verifies that the 47-node radial tree covers the initial box (8). Every leaf is excluded for hypothetical packings with $R < R_0$ by Lemma 3.2 or Lemma 4.1.*

Proof. The replay reconstructs every interval from the root and exact rational child cuts; it never treats the recorded leaf boxes as independent assumptions. Every integer angle tick is checked by (10). Each listed directed negative cycle is checked for consecutive edge endpoints, use of the correct constraints, absence of duplicated vertices, and a strictly negative rational weight sum. At the unique barrier leaf, the checker recomputes all four families of inequalities in Lemma 4.1, and proves $s_9 + 9 > 17$ for every point of that leaf. The checker checks the entire original set of 60 orders at each leaf, and the full binary-tree identity. The resulting ledger is shown in Table 1. $\square$

Table 1: Rational proof-tree replay for disks 4, ..., 9.

Nodes	47
Rational splits	23
Leaves	24
Negative-cycle leaves (all 60 orders)	23
Barrier leaves (59 cycles and 1 analytic barrier)	1
Maximum tree depth	14
Unresolved leaves	0

The minimum rational negative-cycle margin in the checked file is

$$\frac{974034641}{50000000000} > 0.$$

For the barrier leaf, the radial lower endpoints (shown in decimal for readability; exact fractions appear in the certificate) satisfy

Disk i	Certified lower endpoint ℓ_i	Upper endpoint $U - i$
4	13.6748954310675	15.23319390809
5	12.9248954310675	14.23319390809
6	11.1165969540450	13.23319390809
7	10.6165969540450	12.23319390809
8	10.1165969540450	11.23319390809
9	9.1165969540450	10.23319390809

Proof of Theorem 1.1. By Lemmas 3.1, 3.2, and 4.1 and Proposition 4.2, no six disks of radii $4, \dots, 9$ fit in a container of radius $R < R_0$. Hence no packing of all nine disks exists below R_0 . Lemma 2.1 and the explicit nine-disk construction establish feasibility at R_0 , proving $R^* = R_0$. $\square$

5 Reproducibility and limitations

The full replay is available in the accompanying files. Run

```
python3 certify_integer_nine_global.py
```

The dispatcher runs the exact root/upper-bound witness checker and the independent exact rational lower-bound tree checker. The generator `gen_nine_global_tree.py` uses floating point only to *discover* possible angle ticks and branch structure; it is not needed for replay. All decisive verifier tests use exact integers and Python `Fraction`, with outward integer-square-root intervals and rational Taylor/arctangent bounds. The Python implementation and interpretation of the finite certificate remain part of the trust base; the proof has not been independently formalized in a theorem prover or peer reviewed.

The numerical value of the optimum agrees with the best-known nine-disk arrangement listed by Packomania, <https://www.packomania.com/ccin/>. Establishing whether an identical proof has appeared elsewhere requires a separate priority search.